Master Thesis supplement: Joint modelling

"Towards Bayesian joint modelling of ranks and scores in the assessment of grant proposal quality"

Zürich, November 4, 2026

Introduction

This document describes a novel Bayesian approach to jointly model scoring and ranking data using the `Inlajoint` package (?? ref inla joint). We first define and describe the new model. We then apply it on the Gallo et al. (??) dataset and report the results it provides. Finally, we discuss the novel model's benefits and drawbacks in comparison to other joint modelling approaches. During the model fitting, we provide all code necessary to reproduce the analysis performed in this document.

Model definition

To unify the scale of the scoring and ranking data, we first transform the data through z-scoring as follows:

$$y^z_{i,j} = \frac{y_{ij} - \hat{\bar{y}}}{\hat{\sigma}_y}$$

$$r^z_{i,j} = \frac{r_{ij} - \hat{\bar{r}}}{\hat{\sigma}_r}$$

Using the standardised $y^z_{i,j}$ data we modify the Bayesian hierarchical model for scores defined by Heyard et al. (??) in order to obtain:

$$y^z_{i,j} = \beta^y_0 + \theta^y_i + \lambda^y_j$$

$$\theta^y \sim N(0, \sigma^2_{\theta^y}) \qquad \beta^y_0 \sim N(\mu^y, \sigma^2_y)$$

where θ^y_i represent the latent qualities of an object o_i according to the transformed scoring data, λ^y_j represent the assessor-specific difference in strictness and scoring habit, and β^y_0 represents and estimates the mean standardised score of all objects.

Analogously, we create a second model for the transformed ranking data as follows:

$$y^z_{i,j} = \beta^y_0 + \theta^r_i$$
$$\theta^r \sim N(0, \sigma^2_{\theta^r}) \qquad \beta^r_0 \sim N(\mu^r, \sigma^2_r)$$

where θ^r_i represent the latent qualities of an object o_i according to the transformed ranking data, and β^r_0 represents and estimates the mean standardised rank of all objects.

We join these models by deffining a correlation constraint $\text{corr}(\theta^y, \theta^r) = \rho_\theta$ between the random effects of both models, which are in practice the latent quality variables. The `Inlajoint` package does this automatically and fits both models simultaneously taking this correlation into account. This provides the final model:

$$\begin{cases} y^z_{i,j} = \beta^y_0 + \theta^y_i + \lambda^y_j \\ r^z_{i,j} = \beta^r_0 + \theta^r_i \end{cases}$$
$$\theta^y \sim N(0, \sigma^2_{\theta^y}) \qquad \theta^r \sim N(0, \sigma^2_{\theta^r})$$
$$\text{corr}(\theta^y, \theta^r) = \rho_\theta \tag{1}$$
$$\beta^y_0 \sim N(\mu^y, \sigma^2_y) \qquad \beta^y_0 \sim N(\mu^r, \sigma^2_r)$$

Model fitting & Results

In the Gallo et al. (??) dataset, there are 259 scoring observations, but only 49 ranking observations. This provides us with three paths for the model fitting: (1) We simply analyse the data as-is and ignore the missingness of the ranking data. (2) We analyse only the complete cases in this data set and remove 210 scoring observation. (3) We perform weighting to account for the difference in the amount of observations. While path (3) is the most promising for real-world applications, it is beyond the scope of this thesis. We therefore only provide an example of a model fit using path (1) and path (2).

As-is analysis

The code below was used to fit the joint model:

```
# Overview of data
head(dat_as_is)
```

```
  RandomRevID grant_id  std_score std_rank
1         225        1 0.02319127       NA
2         218        1 0.02319127       NA
3         145        1         NA       NA
4         235        1 0.02319127       NA
```

```
5          308         1 0.02319127       NA
6          203         1 0.02319127       NA
```

```
# Fitting model
inla_sr_joint_as_is <- joint(
  formLong = list(
    std_score  ~ 1 + (1 | grant_id) + (1| RandomRevID),
    std_rank  ~ 1 + (1 | grant_id)
  ),
  dataLong = dat_as_is, id = "grant_id",
  family  = c("gaussian", "gaussian"),
  corLong = T
)

# Please check the correlation structure again if you refit model ??
```

The results of the as-is model are the following:

```
cat("TODO ??")
```

```
TODO ??
```

Complete cases

The code below was used to fit the joint model:

```
# Overview of data
head(dat_c_cases)
```

```
   RandomRevID grant_id  std_score   std_rank
27         269        3 -0.6379842  1.4322239
31         225        4 -0.6379842 -0.2929549
32         218        4  0.6625220  0.8571643
36         203        4  1.3127751  0.8571643
37         269        4  1.3127751  0.8571643
38         321        4 -0.6379842  0.8571643
```

```r
# Fitting model
inla_sr_joint_complete <- joint(
  formLong = list(
    std_score  ~ 1 + (1 | grant_id) + (1| RandomRevID),
    std_rank  ~ 1 + (1 | grant_id)
  ),
  dataLong = dat_c_cases, id = "grant_id",
  family  = c("gaussian", "gaussian"),
  corLong = T
)

# Please check the correlation structure again if you refit model ??
```

and provided the following results:

```r
cat("TODO ??")
```

```
TODO ??
```

Discussion and limitations

Advantage: Proof-of-concept that this model can be fitted. Code available for further extensions.

limitations: Check correlation structures, using z-scoring on ranking data is a polarising topic in the literature. Complete case analysis and ignoring missingness both wrong.

Future: Check if combining the data can help in comparison to separate on simulation data where the ground truth is known. Comparison with a simple inla fit just on scores and a simple fit just on ranks.